

MD. AKASH MIAH

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Professional Summary

AI/ML Engineer with hands-on experience in machine learning model development, deployment, and optimization. Skilled in building end-to-end AI systems using Python, PyTorch, and TensorFlow. Strong background in computer vision, LLM-based systems, and data-driven problem solving, with experience translating real-world requirements into production-ready AI solutions.

Technical Skills

Programming & ML: Python, Machine Learning, Deep Learning, Model Training & Evaluation, Optimization

Frameworks: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, OpenCV

AI Techniques: Computer Vision, NLP, Transformers, Fine-tuning, Prompt Engineering

Data Science: EDA, Statistical Analysis, Regression, Model Validation

Tools: Docker, Git, SQL, Google Colab, Python socket programming, Fast API

LLMs & GenAI: LLM inference, LangChain, N8n, RAG, Qdrant, PGvector, Pinecone, Chroma, OpenAI/Gemini/GROQ APIs

Professional Experience

AI Engineer

Betopia LTD

Jul 2025 – Present

Dhaka, Bangladesh

- Contributed to transforming Betopia into an **AI-first platform** through LLM systems and intelligent tooling
- Built and deployed an internal **LLM provider platform**, enabling unified API access to multiple models
- Developed an internal **coding agent**, leveraging proprietary LLMs to automate engineering workflows
- Fine-tuned **transformer-based models** for domain-specific tasks on custom datasets
- Optimized **LLM inference pipelines**, reducing latency and improving response consistency
- Designed **scalable backend systems** for AI services, supporting real-time inference and integrations

Education

B.Sc. in Computer Science and Engineering

North South University

2025

Dhaka, Bangladesh

Professional Development: Short Course on Data Science — Prof. Jennifer Widom, Stanford University (Feb 2024)

Key Projects

AI-Based Grocery Checkout System

Python, OpenCV, YOLO, RT-DETR

- Built an AI-powered checkout system that detects and classifies **10 grocery product classes** from a single image
- Trained YOLOv8 on a custom dataset, achieving **0.995 mAP@50** on the validation set
- Optimized inference pipeline to **696 ms/frame**, enabling near real-time performance
- Designed the system to simulate barcode scanning by mapping detections to **GTIN-based product IDs**
- Engineered the solution for seamless **POS software integration**

Autonomous Chemical Spraying Robot

YOLOv8, IoT, Edge AI

- Developed an autonomous agricultural robot combining **computer vision, IoT, and robotics**
- Deployed YOLOv8 on edge hardware for real-time crop and weed classification
- Integrated perception outputs with robotic actuation to enable **targeted chemical spraying**
- Improved spraying efficiency by minimizing unnecessary chemical usage

RAGbot — AI Knowledge Assistant

LangChain, FAISS, Gemini API

- Implemented a **Retrieval-Augmented Generation (RAG)** pipeline for document-based question answering
- Used FAISS for semantic search over embedded knowledge bases
- Integrated Gemini API to generate accurate, context-aware responses
- Built a desktop GUI to support **enterprise knowledge retrieval and internal documentation search**